Equations and related quantities

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: Solve equations by keeping both sides equal and connect a rule to a table and graph.

R1. Find 140% of 30 points.

R2. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

R3. Find $2.8 \div 0.07$.

R4. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

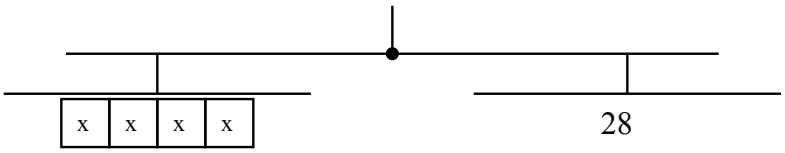
P1. Solve $x + 7 = 19$. Show the same change on both sides.

P2. Four identical bags hold 28 marbles. Write and solve an equation for marbles per bag, b.

P3. Solve $1.2x = 5.4$. Check with multiplication.

P4. Solve $(\frac{2}{3})\,x = 10$. Draw three equal parts of x first.

Lesson model



4 equal bags 28 marbles

Equations and related quantities

Practice and compare methods

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P5. A game costs $18. Mina has $6.50 saved. Write and solve an equation for the extra amount, a, she needs.

P6. Notebooks cost $2.50 each. Use c = 2.5n for n = 0, 1, 2, 4. Plot (n, c). Name the input and output.

F. Complete: 5x = 35. Divide both sides by __. Then x = __.

D. Omar solves 6x = 42 by writing x = 42 − 6 = 36. Explain and repair his work.

M1. 24 shells are 30% of a collection. Find the whole collection.

M2. Find $2 \div \frac{2}{3}$.

M3. Find 3.45 + 5.078.

X1. Solve x + 4 = 7. Show a change to both sides.

Worked example - compare with the slide model

Solve x + 2.75 = 8.

Both sides must stay equal.

Subtract 2.75 from each side: x = 5.25.

Check: 5.25 + 2.75 = 8.

Result: x = 5.25

Equations and related quantities

Challenge, explain and check

20 daily questions include C1 and C2. Use separate paper when helpful.

C1. Three identical snack bags and a $4 delivery fee cost $19. Find the price per bag. A second shop charges $6 per bag with no fee. Which shop is cheaper for five bags, and by how much?

C2. A tank starts with 3 liters and gains 2 liters each minute. Make a table for 0, 1, 3, 5 minutes. Write a rule. When will it hold 17 liters? Explain why multiplying the time by 2 alone is insufficient.

Finish the lesson

S. I can check a solution by __.

E. Solve $(\frac{3}{5})$ x = 12. Show equal parts or an equation step. Check your value.